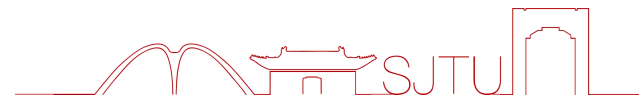




上海交通大学
SHANGHAI JIAO TONG UNIVERSITY



蛋白质结构分析

从序列到结构

2023年11月3日

饮水思源 · 爱国荣校



1

背景/理论基础

2

蛋白质分析流程

3

性质结构分析

01

背景/理论基础





- 
1. 蛋白质的高级结构由其一级结构序列决定，蛋白质可以自发地折叠成它们的天然结构，特别是结构简单的蛋白质小分子。
 2. 序列相似的蛋白质具有相似的三维结构；
 3. 不同的蛋白质中相同的结构域(*domain*)具有相似的功能。



02

蛋白质分析流程

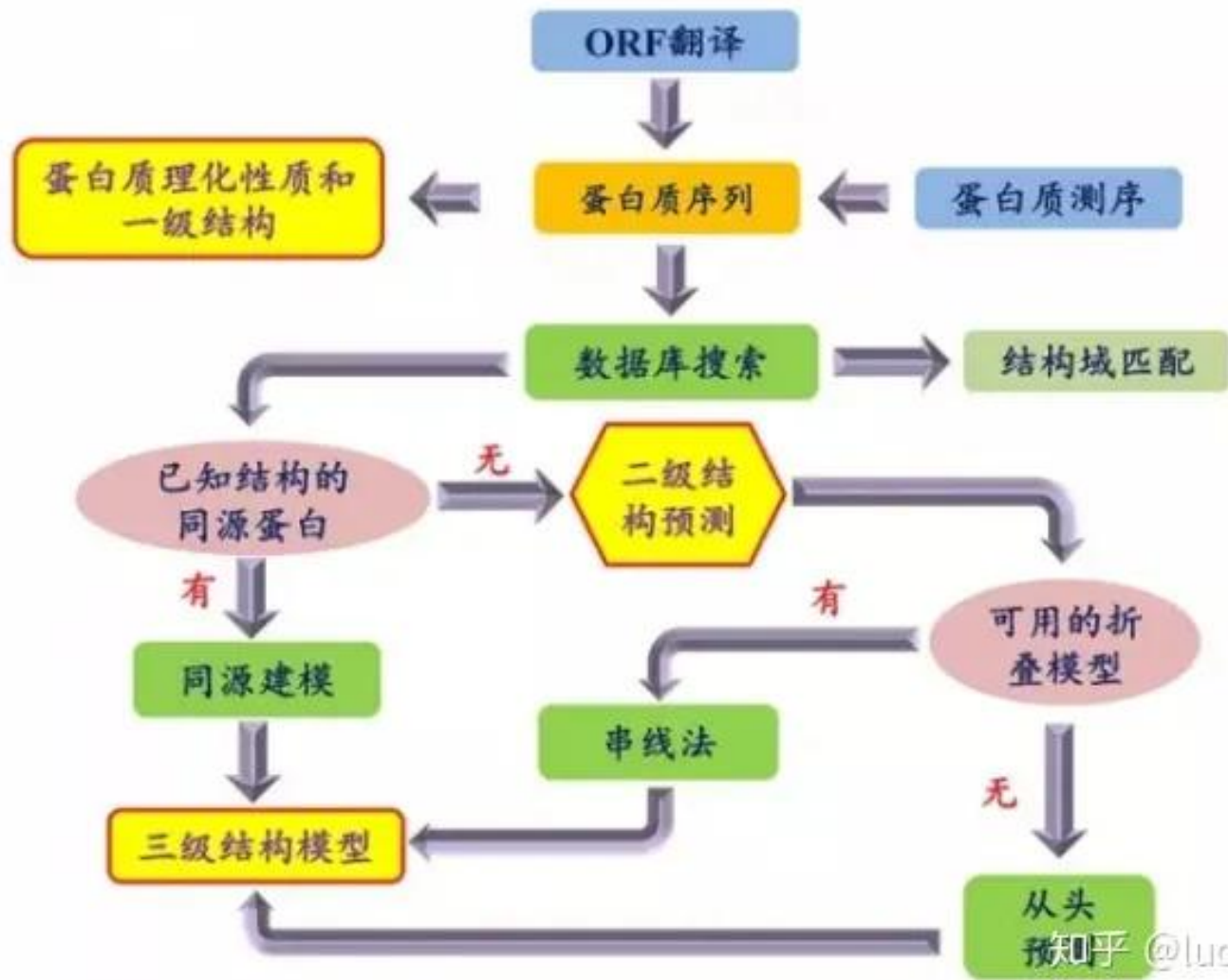




分析流程

一级结构 (1°): 独特的氨基酸序列, 由遗传物质决定。

二级结构 (2°): 多肽链的主链骨架本身 (不包括R基团) 在空间上有规律的折叠和盘绕, 它是由氨基酸残基非侧链基团之间的氢键决定的。



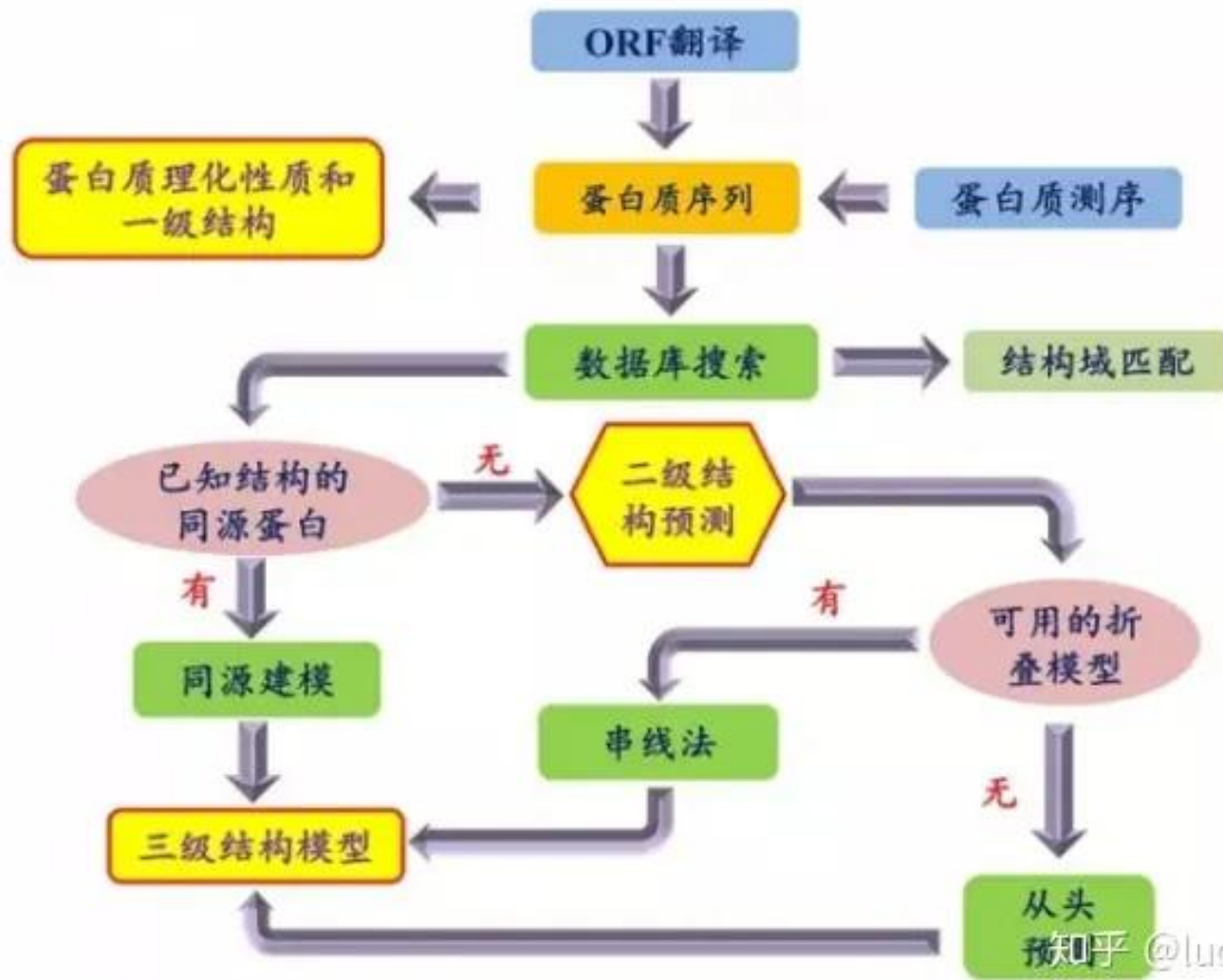


分析流程

三级结构 (3°): 是指多肽链在二级结构的基础上, 进一步盘绕、卷曲和折叠, 形成主要通过氨基酸侧链以次级键以及二硫键维系的完整的三维结构。

四级结构 (4°) 多条肽链

由于新技术的出现, 建立新的链路



03

性质结构分析





理化性质

Expasy - ProtParam tool

pI 、 Mw 、氨基酸组成、消光系数、稳定系数、亲水性

Expasy

ProtParam

Home | Contact

ProtParam tool

ProtParam ([References](#) / [Documentation](#)) is a tool which allows the computation of various physical and chemical parameters for a given protein stored in [Swiss-Prot](#) or [TrEMBL](#) or for a user entered protein sequence. The computed parameters include the molecular weight, theoretical pI , amino acid composition, atomic composition, extinction coefficient, estimated half-life, instability index, aliphatic index and grand average of hydropathicity (GRAVY) ([Disclaimer](#)).

Please note that you may only fill out **one** of the following fields at a time.

Enter a Swiss-Prot/TrEMBL accession number (AC) (for example **P05130**) or a sequence identifier (ID) (for example **KPC1_DROME**):

Or you can paste your own amino acid sequence (in one-letter code) in the box below:

RESET

Compute parameters

Expasy is operated by the SIB Swiss Institute of Bioinformatics | [Terms of Use](#)

[Back to the top](#)

”

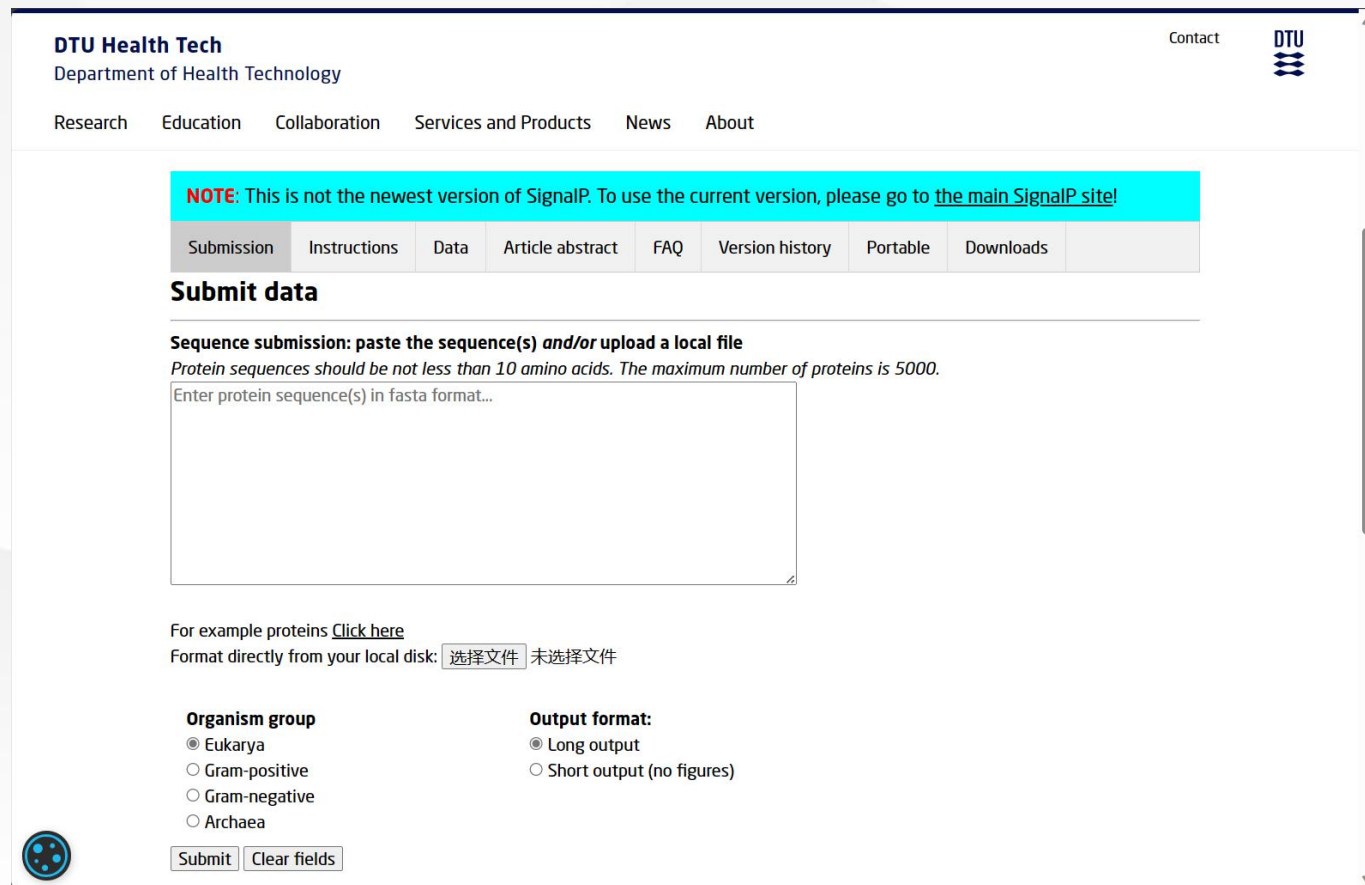




信号肽预测 SignalP

SignalP 5.0 - DTU Health Tech - Bioinformatic Services

是引导新合成的蛋白质向分泌通路转移的短（长度5-30个氨基酸）肽链。



The screenshot shows the SignalP 5.0 web interface. At the top, there is a header for "DTU Health Tech" (Department of Health Technology) with navigation links: Research, Education, Collaboration, Services and Products, News, and About. A "Contact" link and the DTU logo are also present. A prominent red banner states: "NOTE: This is not the newest version of SignalP. To use the current version, please go to [the main SignalP site!](#)". Below this is a navigation bar with tabs: Submission, Instructions, Data, Article abstract, FAQ, Version history, Portable, and Downloads. The "Submit data" section is active, showing instructions: "Sequence submission: paste the sequence(s) and/or upload a local file" and "Protein sequences should be not less than 10 amino acids. The maximum number of proteins is 5000." A text area is provided for "Enter protein sequence(s) in fasta format...". Below the text area, there are links for "For example proteins" and "Click here". A section for "Format directly from your local disk:" includes buttons for "选择文件" (Select file) and "未选择文件" (No file selected). The "Organism group" section has radio buttons for Eukarya (selected), Gram-positive, Gram-negative, and Archaea. The "Output format:" section has radio buttons for Long output (selected) and Short output (no figures). At the bottom, there are "Submit" and "Clear fields" buttons. A small circular icon with four dots is visible in the bottom left corner of the interface.





PSIPRED

[PSIPRED Workbench \(ucl.ac.uk\)](http://ucl.ac.uk)

二级结构预测

- α 螺旋 (alpha-helix) 及其他螺旋
- β 折叠 (beta-sheet)
- β 转角 (beta-turn)
- β 突起 (beta-bulge)
- 环 (loop) 与无规则卷曲 (random coil)

PSIPRED

UCL Department of Computer Science: Bioinformatics Group

UCL

MAIN NAVIGATION

- Introduction
- Contact
- Downloads & Branding
- Twitter/News

PSIPRED Team Links

- People
- ProCovar
- Publications
- Vacancies

PSIPRED Workbench Links

- PSIPRED Workbench
- Workbench Overview
- Workbench Citation
- Help & Tutorials
- REST API
- PSIPRED Github

The PSIPRED Workbench provides a range of protein structure prediction methods. The site can be used interactively via a web browser or programmatically via our REST API. For high-throughput analyses, downloads of all the algorithms are available.

Amino acid sequences enable: secondary structure prediction, including regions of disorder and transmembrane helix packing; contact analysis; fold recognition; structure modelling; and prediction of domains and function. In addition **PDB Structure files** allow prediction of protein-metal ion contacts, protein-protein hotspot residues, and membrane protein orientation.

Data Input

Select input data type

☒ Sequence Data ☐ PDB Structure Data

Choose prediction methods (hover for short description)

Popular Analyses

☒ PSIPRED 4.0 (Predict Secondary Structure) ☐ DISOPRED3 (Disopred Prediction)

☐ MEMSAT-SVM (Membrane Helix Prediction) ☐ pGenTHREADER (Profile Based Fold Recognition)

Contact Analysis

☐ DeepMetaPSICOV 1.0 (Structural Contact Prediction) ☐ MEMPACK (TM Topology and Helix Packing)

Fold Recognition

☐ GenTHREADER (Rapid Fold Recognition) ☐ pDomTHREADER (Protein Domain Fold Recognition)

Structure Modelling

☐ Bioserf 2.0 (Automated Homology Modelling) ☐ Domserf 2.1 (Automated Domain Homology Modelling)

☐ DMPfold 1.0 Fast Mode (Protein Structure Prediction)

Single Sequence Prediction

☐ S4Pred 1.2 (Single Sequence SS prediction)

Required Options

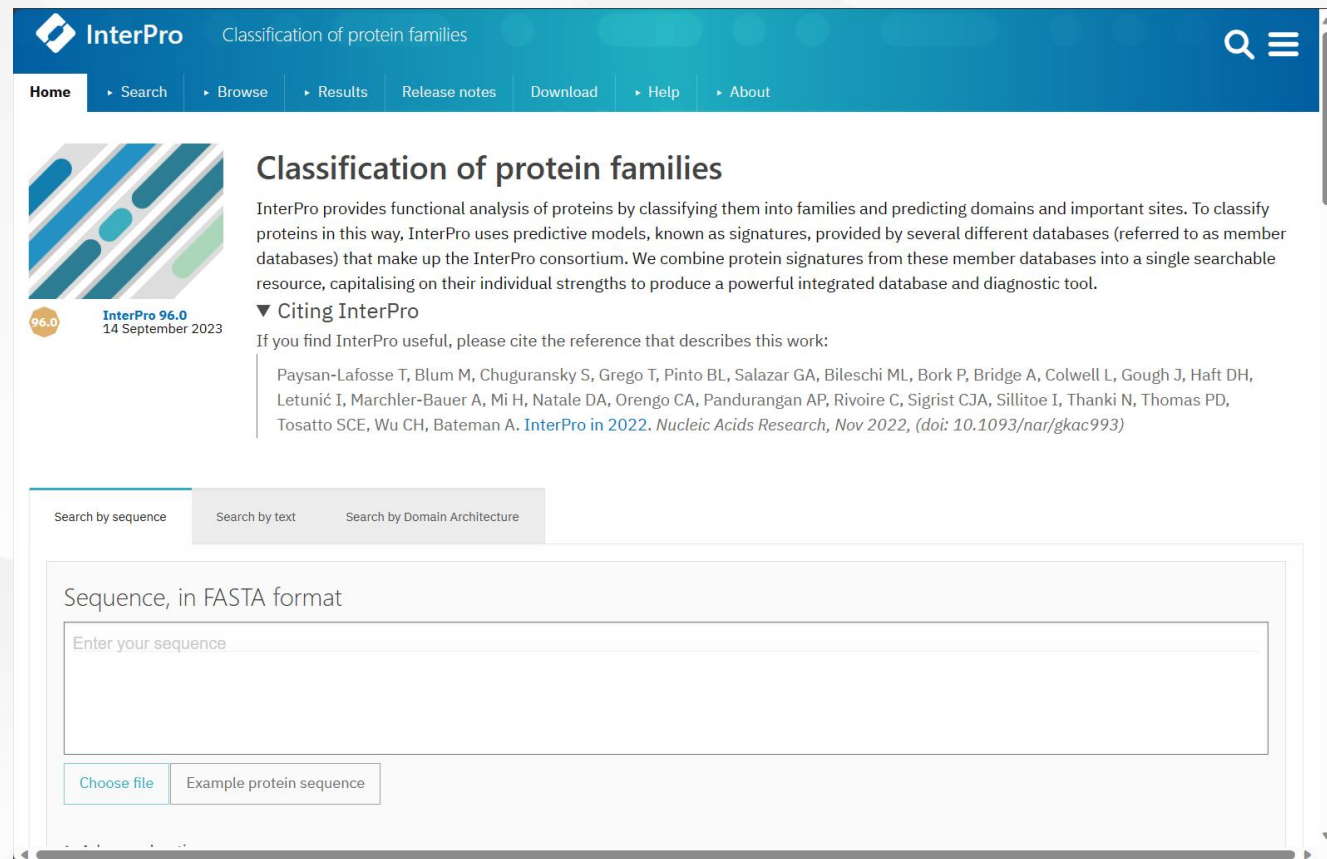




InterPro

[InterPro \(ebi.ac.uk\)](http://ebi.ac.uk)

是在一个蛋白质分子内的相对独立的**球状结构**和/或**功能模块**，是由若干个结构模体组成的相对独立的球形结构单位，它们通常是独自折叠形成的，**与蛋白质的功能直接相关**。某些结构域在同一个蛋白质分子上被重复使用，某些蛋白质由多个拷贝的一种和多种结构域组成。



The screenshot shows the InterPro website, titled "Classification of protein families". The navigation bar includes links for Home, Search, Browse, Results, Release notes, Download, Help, and About. The main content area features a logo with the text "InterPro 96.0 14 September 2023". Below the logo, the title "Classification of protein families" is followed by a description: "InterPro provides functional analysis of proteins by classifying them into families and predicting domains and important sites. To classify proteins in this way, InterPro uses predictive models, known as signatures, provided by several different databases (referred to as member databases) that make up the InterPro consortium. We combine protein signatures from these member databases into a single searchable resource, capitalising on their individual strengths to produce a powerful integrated database and diagnostic tool." A section titled "Citing InterPro" provides a reference: "If you find InterPro useful, please cite the reference that describes this work: Paysan-Lafosse T, Blum M, Chuguransky S, Grego T, Pinto BL, Salazar GA, Bileschi ML, Bork P, Bridge A, Colwell L, Gough J, Haft DH, Letunic I, Marchler-Bauer A, Mi H, Natale DA, Orengo CA, Pandurangan AP, Rivoire C, Sigrist CJA, Sillitoe I, Thanki N, Thomas PD, Tosatto SCE, Wu CH, Bateman A. *InterPro in 2022. Nucleic Acids Research, Nov 2022, (doi: 10.1093/nar/gkac993)*". At the bottom, there are three search tabs: "Search by sequence", "Search by text", and "Search by Domain Architecture". The "Search by sequence" tab is active, showing a text input field labeled "Sequence, in FASTA format" with the placeholder "Enter your sequence". Below the input field are two buttons: "Choose file" and "Example protein sequence".



方法一：同源建模

[SWISS-MODEL Interactive Workspace](https://expasy.org)
(expasy.org)

蛋白质结构同源模建「又称同源模拟、同源建模」的理论基础是蛋白质的三级结构比蛋白质的一级结构更为保守。





方法二：折叠识别

「pGenThreader」

[PSIPRED Workbench \(ucl.ac.uk\)](http://PSIPRED.Workbench(ucl.ac.uk))

基本原理：从蛋白质结构数据库中识别与待测序列具有相似折叠类型，进而实现对待测序列的空间结构预测。

PSIPRED UCL Department of Computer Science: Bioinformatics Group

MAIN NAVIGATION

- Introduction
- Contact
- Downloads & Branding
- Twitter/News

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- People
- ProCovar
- Publications
- Vacancies

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Required Options





方法三：从头计算法¹

[TASSER server for protein structure and function prediction \(zhanggroup.org\)](http://TASSER.server.for.protein.structure.and.function.prediction(zhanggroup.org))

原理：蛋白质的天然构象对应其能量最低的构象「热力学理论」，因此通过构造合适的能量函数及优化方法，实现从蛋白质序列直接预测其三维结构的目的。



The screenshot shows the I-TASSER website. At the top, there's a navigation bar with links: Home, Research, COVID-19, Services, Publications, People, Teaching, Job Opening, News, Forum, and Lab Only. The main header features the 'Zhang Lab' logo and the 'UNIVERSITY OF MICHIGAN' logo. Below the navigation bar, the 'Online Services' menu is visible on the left, listing various tools like I-TASSER, I-TASSER-MTD, C-I-TASSER, CR-I-TASSER, QUARK, C-QUARK, LOMETS, MUSTER, CETHREADER, SEGMENT, DeepFold, DeepFoldRNA, FoldDesign, COFACTOR, COACH, MetaGO, TripletGO, IonCom, FG-MD, ModRefiner, REMO, DEMO, DEMO-EM, SPRING, COTH, and Threpp. The main content area displays the 'I-TASSER Protein Structure & Function Predictions' logo and a brief description of the server's capabilities and history. It mentions that the server has completed predictions for 75,579 proteins submitted by 18,702 users from 161 countries, and the template library was updated on 2023/10/12. Below this, there's a section for 'I-TASSER On-line Server' with a form to input a protein sequence (FASTA format) and a button to 'View an example of I-TASSER output'. There are also links for 'Queue', 'Forum', 'Download', 'Search', 'Registration', 'Statistics', 'Remove', 'Potential', 'Decoys', 'News', 'Annotation', 'About', and 'FAQ'. At the bottom, there's a section for 'Or upload the sequence from your local computer' with a '选择文件' (Select File) button and a '未选择文件' (No file selected) status. Below this, there's a field for 'Email: (mandatory, where results will be sent to)' and a 'Password: (mandatory, please click here if you do not have a password)' field.





方法四：AI预测

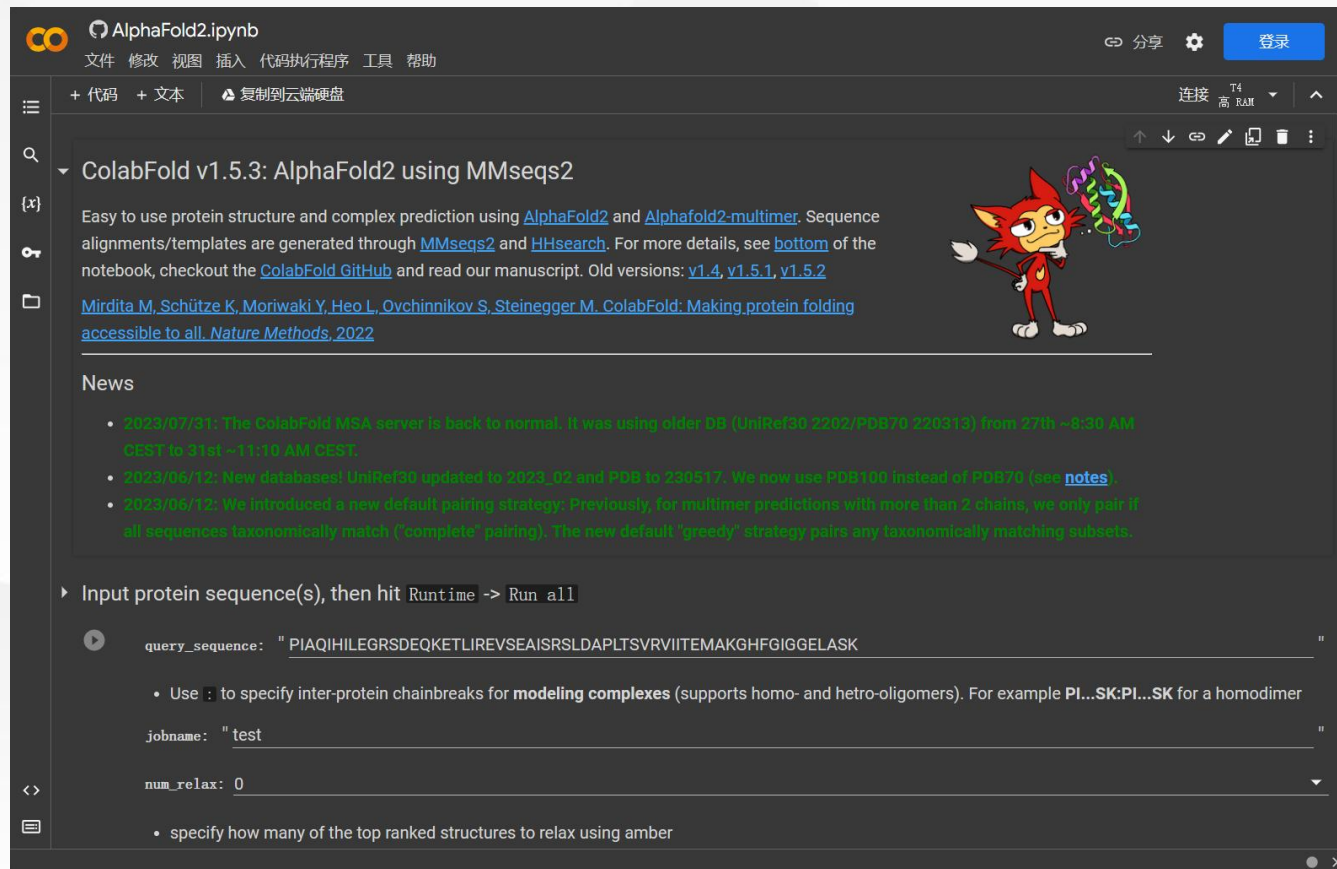
「AlphaFold」

[AlphaFold2.ipynb - Colaboratory](https://colab.research.google.com/github/google/alphafold2/blob/master/AlphaFold2.ipynb)

[google.com](https://www.google.com)

[AlphaFold Protein Structure](https://alphafold.ebi.ac.uk/)

[Database](https://alphafold.ebi.ac.uk/)





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感谢聆听

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